

$$\textcircled{2} \quad \frac{d}{dt} X = \begin{pmatrix} -a & 1 \\ -b & 1 \end{pmatrix} X$$

$$\tau = 1 - a$$

$$\det = -a + b$$

$$\Delta = (1 - a)^2 - 4(-a + b)$$

$$= 1 - 2a + a^2 + 4a - 4b$$

$$= a^2 + 2a - 4b + 1$$

Case silla

$$\det < 0 \quad \begin{cases} b = 0 & \vee & a \neq 0 \\ b < 0 & \vee & a > 0 \end{cases}$$

$$\begin{aligned} -a + b &= 0 \\ b &= a \end{aligned}$$

Foco Estable

$$\gamma < 0 \rightarrow 1 - a < 0 \rightarrow a > 1$$

$$d > 0 \rightarrow -a + b > 0 \rightarrow a > 1 \text{ y } b > a$$

$$\Delta < 0 \rightarrow a^2 + 2a - 4b + 1 < 0$$

$$\rightarrow a > 1 \text{ y } b > a$$

Foco Inestable

$$\gamma > 0 \rightarrow 1 - a > 0 \rightarrow a < 1$$

$$det > 0 \rightarrow -a + b > 0 \rightarrow b > a \rightarrow a < 1 \text{ y } b > a$$

$$\Delta < 0 \rightarrow 1 + 2a + a^2 - 4b < 0$$

Nodo Estable

$$\gamma < 0 \rightarrow 1 - a < 0 \rightarrow a < 1$$

$$d > 0 \rightarrow -a + b > 0 \rightarrow b > a$$

$$\Delta > 0 \rightarrow a^2 + 2a - 4b + 1 > 0$$

$$a^2 + 2a - 4b + 1 = 0$$

$$-4b = -a^2 - 2a - 1$$

$$b = \frac{a^2 + 2a + 1}{4}$$

$$b = \frac{a^2}{4} + \frac{a}{2} + \frac{1}{4}$$

Nº de Installs

$$x > 0 \rightarrow 1 - a > 0 \rightarrow 1 > a$$

$$\det > 0 \rightarrow -a + b > 0 \rightarrow b > a$$

$$\Delta > 0 \rightarrow a^2 + 2a - 4b + 1 > 0$$

$$a^2 + 2a - 4b + 1 = 0$$

$$b = \frac{a^2}{4} + \frac{a}{2} + \frac{1}{4}$$

Nº de Impropriedades:

Encima de la parábola

